6. The top of a 15 m high tower makes an angle of elevation of 60° with the bottom of an electronic pole and angle of elevation of 30° with the top of the pole. What is the height of the electric pole?

6.3.2 Problems involving Angle of Depression

In this section, we try to solve problems when Angles of depression are given.

Note

Angle of Depression and Angle of Elevation are equal because they are alternative angles.

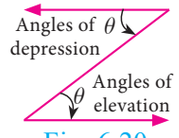


Fig. 6.20

Example 6.26 A player sitting on the top of a tower of height 20 m observes the angle of depression of a ball lying on the ground as 60°. Find the distance between the foot of the tower and the ball. ($\sqrt{3} = 1.732$)

Solution Let BC be the height of the tower and A be the position of the ball lying on the ground. Then,

BC = 20 m and $\angle XCA = 60° = \angle CAB$

Let AB = x metres.

In the right angled ΔABC,

$$\tan 60° = \frac{BC}{AB}$$

$$\sqrt{3} = \frac{20}{x}$$

$$x = \frac{20 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{20 \times 1.732}{3} = 11.55 \text{ m.}$$

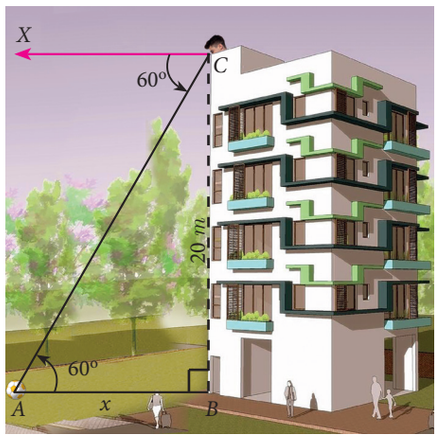


Fig. 6.21

Hence, the distance between the foot of the tower and the ball is 11.55 m.

Example 6.27 The horizontal distance between two buildings is 140 m. The angle of depression of the top of the first building when seen from the top of the second building is 30°. If the height of the first building is 60 m, find the height of the second building. ($\sqrt{3} = 1.732$)

Solution The height of the first building AB = 60 m. Now, AB = MD = 60 m

Let the height of the second building CD = h. Distance BD = 140 m

Now, AM = BD = 140 m

From the diagram,

$\angle XCA = 30° = \angle CAM$

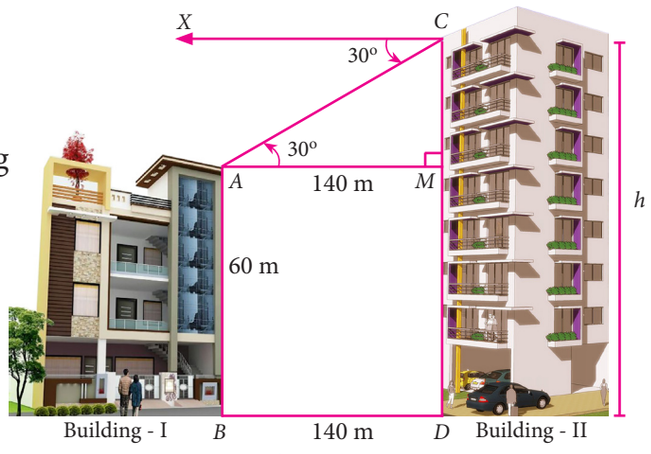


Fig. 6.22

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